LECTURE 5
SIMULTANEOUS EQUATIONS IV: LIMITED INFORMATION ML (LIML)

In this lecture, we consider ML estimation of a single equation that is part of a system of simultaneous equations. Without loss of generality, we can focus on the first equation:

$$\begin{aligned} y_{1i} &= X_{1,i}^{\top}\delta_1 + u_{1i} \\ &= Y_{1,i}^{\top}\gamma_1 + Z_{1,i}^{\top}\beta_1 + u_{1i}, \end{aligned}$$

where $Y_{1,i}$ and $Z_{1,i}$ are the vectors of included endogenous and exogenous regressors respectively, as defined in Lecture 2. For the included endogenous regressors we have the following reduced form equation

$$Y_{1,i} = \Pi_1 Z_{1,i} + \Pi_2 Z_{2,i} + V_{1,i}.$$

We ignore $Y_{1,i}^{*}$, the vector of endogenous variables excluded from the first equation. The two equations above can be written together as

$$\begin{pmatrix} 1 & -\gamma_1^{\top} \\ 0 & I_{m_1} \end{pmatrix} \begin{pmatrix} y_{1i} \\ Y_{1,i} \end{pmatrix} = \begin{pmatrix} \beta_1^{\top} & 0 \\ \Pi_1 & \Pi_2 \end{pmatrix} \begin{pmatrix} Z_{1,i} \\ Z_{2,i} \end{pmatrix} + \begin{pmatrix} u_{1i} \\ V_{1,i} \end{pmatrix},$$

or

$$\widetilde{\Gamma}_1 \widetilde{Y}_{1,i} = \widetilde{B}_1 Z_i + \widetilde{U}_i,$$

where

$$\begin{aligned} \widetilde{\Gamma}_1 &= \begin{pmatrix} 1 & -\gamma_1^{\top} \\ 0 & I_{m_1} \end{pmatrix}, \\ \widetilde{B}_1 &= \begin{pmatrix} \beta_1^{\top} & 0 \\ \Pi_1 & \Pi_2 \end{pmatrix}, \\ \widetilde{Y}_{1,i} &= \begin{pmatrix} y_{1i} \\ Y_{1,i} \end{pmatrix}, \\ \widetilde{U}_i &= \begin{pmatrix} u_{1i} \\ V_{1,i} \end{pmatrix}. \end{aligned}$$

Assuming that

$$\widetilde{U}_i | Z_i \sim N\left(0, \widetilde{\Sigma}_1\right),$$

similarly to the derivation of equation (3) in Lecture 4, we obtain that the concentrated log-likelihood for $\widetilde{Y}_{1,i}$ is

$$\begin{aligned} Q_n\left(\widetilde{\Gamma}_1, \widetilde{B}_1\right) &= -\frac{(m_1+1)}{2}\left(\log(2\pi)+1\right) + \log\left|\widetilde{\Gamma}_1\right| - \frac{1}{2}\log\left| n^{-1}\sum_{i=1}^{n}\left(\widetilde{\Gamma}_1\widetilde{Y}_{1,i} - \widetilde{B}_1 Z_i\right)\left(\widetilde{\Gamma}_1\widetilde{Y}_{1,i} - \widetilde{B}_1 Z_i\right)^{\top}\right| \\ &= -\frac{(m_1+1)}{2}\left(\log(2\pi)+1\right) - \frac{1}{2}\log\left| n^{-1}\sum_{i=1}^{n}\left(\widetilde{\Gamma}_1\widetilde{Y}_{1,i} - \widetilde{B}_1 Z_i\right)\left(\widetilde{\Gamma}_1\widetilde{Y}_{1,i} - \widetilde{B}_1 Z_i\right)^{\top}\right|, \end{aligned}$$

where the last equality follows from the fact that, due to the restricted structure of $\widetilde{\Gamma}_1$, $\left|\widetilde{\Gamma}_1\right| = 1$.

Thus, LIML is a special case of FIML with properly defined matrices of parameters. However, because of the restricted structure of $\widetilde{\Gamma}_1$, there is a closed-form expression for the LIML estimator. Let $\widehat{\delta}_1$ be the LIML estimator of $\delta_1 = \left(\gamma_1^{\top}, \beta_1^{\top}\right)^{\top}$, then using the matrix notation of Lecture 2, we can write

$$\widehat{\delta}_1 = \left(X_1^{\top}\left(I_n - \lambda M\right)X_1\right)^{-1} X_1^{\top}\left(I_n - \lambda M\right)y_1,$$

where

$$\begin{aligned} M &= I_n - P, \\ P &= Z\left(Z^\top Z\right)^{-1} Z^\top, \end{aligned}$$

and P is the projection matrix onto the space spanned by the exogenous variables Z_i (included and excluded from the first equation),

$$\lambda = \min_t \frac{t^\top W_1 t}{t^\top W t},$$

where

$$\begin{aligned} W &= \begin{pmatrix} y_1 & Y_1 \end{pmatrix}^\top M \begin{pmatrix} y_1 & Y_1 \end{pmatrix}, \\ W_1 &= \begin{pmatrix} y_1 & Y_1 \end{pmatrix}^\top M_1 \begin{pmatrix} y_1 & Y_1 \end{pmatrix}, \end{aligned}$$

and

$$\begin{aligned} M_1 &= I_n - P_1, \\ P_1 &= Z_1\left(Z_1^\top Z_1\right)^{-1} Z_1^\top, \end{aligned}$$

and M_1 is the projection matrix onto the space orthogonal to that spanned by $Z_{1,i}$, the exogenous variables included in the first equation. (As defined above, λ is actually the smallest eigenvalue of $W_1 W^{-1}$.)

Next, we will show the asymptotic equivalence of LIML and 2SLS estimators. First, we will show that $\lambda \geq 1$.

$$\begin{aligned} t^\top W_1 t - t^\top W t &= t^\top \begin{pmatrix} y_1 & Y_1 \end{pmatrix}^\top (M_1 - M) \begin{pmatrix} y_1 & Y_1 \end{pmatrix} t \\ &= t^\top \begin{pmatrix} y_1 & Y_1 \end{pmatrix}^\top (P - P_1) \begin{pmatrix} y_1 & Y_1 \end{pmatrix} t. \end{aligned}$$

Since Z_1 is a part of Z, $PZ_1 = Z_1$, and, therefore, $PP_1 = P_1$. Hence,

$$\begin{aligned} (P - P_1)(P - P_1) &= P - P_1 P - P P_1 + P_1 \\ &= P - P_1, \end{aligned}$$

idempotent and, therefore, positive semidefinite. Thus, $t^\top W_1 t - t^\top W t \geq 0$ for any t and $\lambda \geq 1$.

Next, define

$$u_1 = \begin{pmatrix} u_{1i} \\ \vdots \\ u_{1n} \end{pmatrix}.$$

We have

$$\begin{aligned} & \min_t \frac{t^\top W_1 t}{t^\top W t} \\ = \; & \frac{\begin{pmatrix} 1 & -\gamma_1^\top \end{pmatrix} W_1 \begin{pmatrix} 1 & -\gamma_1^\top \end{pmatrix}^\top}{\begin{pmatrix} 1 & -\gamma_1^\top \end{pmatrix} W \begin{pmatrix} 1 & -\gamma_1^\top \end{pmatrix}^\top} \\ \leq \; & \frac{\begin{pmatrix} 1 & -\gamma_1^\top \end{pmatrix} \begin{pmatrix} y_1 & Y_1 \end{pmatrix}^\top M_1 \begin{pmatrix} y_1 & Y_1 \end{pmatrix} \begin{pmatrix} 1 & -\gamma_1^\top \end{pmatrix}^\top}{\begin{pmatrix} 1 & -\gamma_1^\top \end{pmatrix} \begin{pmatrix} y_1 & Y_1 \end{pmatrix}^\top M \begin{pmatrix} y_1 & Y_1 \end{pmatrix} \begin{pmatrix} 1 & -\gamma_1^\top \end{pmatrix}^\top} \\ = \; & \frac{(Z_1\beta_1 + u_1)^\top M_1 (Z_1\beta_1 + u_1)}{(Z_1\beta_1 + u_1)^\top M (Z_1\beta_1 + u_1)} \\ = \; & \frac{u_1^\top M_1 u_1}{u_1^\top M u_1}. \end{aligned}$$

Thus,

$$0 \leq \lambda - 1 \leq \frac{u_1^\top (M_1 - M) u_1}{u_1^\top M u_1} = \frac{u_1^\top (P - P_1) u_1}{u_1^\top M u_1}.$$

Lastly,

$$\begin{aligned} n^{1/2} \frac{u_1^\top (P - P_1) u_1}{u_1^\top M u_1} &= \frac{\frac{u_1^\top Z}{n^{1/2}} \left(\frac{Z^\top Z}{n}\right)^{-1} \frac{Z^\top u_1}{n} - \frac{u_1^\top Z_1}{n^{1/2}} \left(\frac{Z_1^\top Z_1}{n}\right)^{-1} \frac{Z_1^\top u_1}{n}}{\frac{u_1^\top u_1}{n} - \frac{u_1^\top Z}{n} \left(\frac{Z^\top Z}{n}\right)^{-1} \frac{Z^\top u_1}{n}} \\ &\to_p 0, \end{aligned}$$

and, therefore,

$$n^{1/2} (\lambda - 1) \to_p 0.$$

Next,

$$\begin{aligned} I_n - \lambda M &= I_n - \lambda M + M - M \\ &= I_n - M - (\lambda - 1) M \\ &= P - (\lambda - 1) M. \end{aligned}$$

Hence, the difference between the LIML and 2SLS estimators is given by

$$\begin{aligned} n^{1/2} \left(\widehat{\delta}_1 - \widehat{\delta}_1^{2SLS}\right) &= \left(\frac{X_1^\top (I_n - \lambda M) X_1}{n}\right)^{-1} \frac{X_1^\top (I_n - \lambda M) u_1}{n^{1/2}} - \left(\frac{X_1^\top P X_1}{n}\right)^{-1} \frac{X_1^\top P u_1}{n^{1/2}} \\ &= \left(\frac{X_1^\top P X_1 - (\lambda - 1) X_1^\top M X_1}{n}\right)^{-1} \frac{X_1^\top P u_1 - (\lambda - 1) X_1^\top M u_1}{n^{1/2}} - \left(\frac{X_1^\top P X_1}{n}\right)^{-1} \frac{X_1^\top P u_1}{n^{1/2}} \\ &= \left(\left(\frac{X_1^\top P X_1 - (\lambda - 1) X_1^\top M X_1}{n}\right)^{-1} - \left(\frac{X_1^\top P X_1}{n}\right)^{-1}\right) \frac{X_1^\top P u_1}{n^{1/2}} \\ &- \left(\frac{X_1^\top P X_1 - (\lambda - 1) X_1^\top M X_1}{n}\right)^{-1} \frac{(\lambda - 1) X_1^\top M u_1}{n^{1/2}} \\ &\to_p 0. \end{aligned}$$